

One **FUZE** dry-fog pass beats spraying wet disinfectant every cycle.

Our proprietary backpack applicator lays down a sub-5-micron fog of FUZE's durable, non-leaching antimicrobial — full coverage, no wet surfaces, no daily re-spray.



FUZE proprietary build & design

<5µm

DROPLET SIZE

16 L

TANK CAPACITY

8 hrs

RUNTIME / 2AH
PACK

20 V

DEWALT-STYLE
BATTERY

● THE HARDWARE — DRY FOG VS. ELECTROSTATIC SPRAYER

	FUZE DRY-FOG APPLICATOR	TYPICAL ELECTROSTATIC SPRAYER
Droplet size	Under 5 microns — true dry fog; coats evenly and flashes off without wetting.	40–160 microns — 10–30× larger. E-Mist EPIX360 = 75µm; EvaClean Protexus = 40/60/80µm; Ryobi = 50/75/100µm.
Surface result	Stays dry — no pooling, streaks, or wipe-down.	Charged droplets wrap the surface but still leave it wet — visible deposit, needs dwell/dry time.
Power	20V Dewalt-style battery , adapter for 18V Milwaukee M18 packs.	Proprietary pack (e.g. 16.8V Protexus, 12V Electro-Mist).
Runtime	8 hours on a single 2Ah pack (~5W draw). All-day cordless.	~4 hours typical (EvaClean Protexus, 16.8V).
Tank & control	16 L tank , adjustable pressure with gauge + digital readout, pro Kobalt spray gun.	Smaller tanks (1–4 L handheld; backpacks vary); fixed nozzle steps.
Battery flexibility	Runs the platforms you already own .	Locked to the maker's proprietary battery.

● THE CHEMISTRY — FUZE METAMATERIAL VS. SPRAYED DISINFECTANTS

	FUZE METAMATERIAL	QUATS / BLEACH / PEROXIDE / ALCOHOL
How it works	Non-leaching contact-kill. Metamaterial bonds permanently to the surface and dismantles microbes on contact.	Wet chemical kill — active only while the surface stays wet; works by leaching ions / oxidizing.
Durability	Durable, bonded protection that keeps working between cleanings.	Evaporates — protection ends when the surface dries. No meaningful residual.
Re-application	Apply once — long service life per treatment.	Re-spray every cleaning cycle, often multiple times a day.
Dwell time	No wet dwell required — always on.	Quats/bleach need a wet contact window (often up to ~10 min) to be effective.
Resistance risk	Physical kill mechanism — no resistance pathway .	Quaternary ammonium tolerance/resistance is documented in the literature.
Corrosion	Non-corrosive. Trace metamaterial (20 ppm), no chlorine or oxidizers — safe on metals, electronics, seals.	Bleach pits stainless & eats aluminum; peroxide/peracetic attack brass & copper; chloride quats corrode contacts — and degrade the sprayer's own internals.
Surface feel	No change to hand or finish — no film, no residue, no stiffening on textiles.	Quats leave a tacky, dust-grabbing film; bleach leaves chalky residue and can dull finishes; repeated wet passes re-deposit every cycle.
Chemistry & safety	PFAS-free , no binders, no harsh chemistry, no corrosive residue.	Quats & bleach are skin/respiratory irritants and asthma triggers; bleach corrodes; aquatic toxicity.
Validation	Third-party tested — ASTM E2149, AATCC 100, ISO 20743 ; OEKO-TEX Standard 100 Class I, bluesign® approved, EPA registered.	Performance ends at dry-down; no durable, test-backed residual claim.

The combined advantage: the FUZE dry-fog applicator lays down a sub-5-micron, even coat of a durable, non-leaching antimicrobial that bonds to the surface — so instead of re-spraying wet quat or bleach every cleaning cycle (toxic, corrosive, gone at dry-down), you treat once and stay protected between cleanings.

Competitor figures are typical published manufacturer specs (E-Mist, EvaClean Protexus, Ryobi) and general chemistry-class characteristics, shown for comparison. FUZE certifications: OEKO-TEX Standard 100 Class I, bluesign® approved, EPA registered (federal), PFAS-free. Public-facing antimicrobial claims should follow EPA Treated Article guidance — e.g. "inhibits the growth of odor-causing bacteria." Specific organism kill-rate claims require product-specific public-health pesticide registration.